The documentation and process conversion measures necessary to comply with this document shall be completed by 25 April 2021.

INCH-POUND
MIL-PRF-19500/559M
25 January 2021
SUPERSEDING
MIL-PRF-19500/559L
30 July 2015

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, UNITIZED, NPN, SILICON, SWITCHING,
FOUR TRANSISTOR ARRAY, TYPES 2N6989, AND 2N6990,
JAN, JANTX, JANTXV, JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, switching transistors in a four independent chip array. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Provisions for radiation hardness assurance (RHA) to eight radiation levels is provided for JANS product assurance levels. RHA level designators “M”, “D”, “P“, “L”, “R”, “F’, “G”, and “H” are appended to the device prefix to identify devices, which have passed RHA requirements.

1.2 Package outlines. The device packages for the encapsulated device types are as follows: 14-pin dual-in-line (2N6989) in accordance with figure 1, 14-pin flat-pack (2N6990) in accordance with figure 2, and 20-pin leadless chip carrier (2N6989U in accordance with figure 4 and figure 5.

1.3 Maximum ratings unless otherwise specified T_A = +25°C. (1)

Type	P_T T_A = +25°C (2)	P_T $T_{A(AM)}$ = +25°C (2)	$R_{\theta JA}$ (3)	$R_{\theta JA(AM)}$ (3)	V_{CBO} (4)	V_{EBO} (4)	V_{CEO} (4)	I_C	T_J and T_{STG}
	W	W	°C/W	°C/W	V dc	V dc	V dc	mA dc	°C
2N6989	1.5	N/A	85	N/A	75	6	50	(2) 800	-65 To +200
2N6989U	1.0	N/A	160	N/A	75	6	50	(2) 800	
2N6990	1.0	N/A	175	N/A	75	6	50	(2) 800	
2N6990	N/A	1.0	N/A	23	75	6	50	(2)800	

(1) Maximum voltage between transistors shall be ≥ 500 V dc.
(2) For derating, see figure 6, figure 7, figure 8, and figure 9. Ratings apply to total package.
(3) For thermal impedance curves, see figure 10, figure 11, figure 12, and figure 13.
(4) Ratings apply to each transistor in the array.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FSC 5961

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1.4 Primary electrical characteristics. Characteristics apply to each transistor in the array.

Limits	h_{FE2} (1) V_{CE} = 10 V dc I_C = 1.0 mA dc	h_{FE4} (1) V_{CE} = 10 V dc I_C = 150 mA dc	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	Switching	
				t_{on} (see figure 14)	t_{off} (see figure 15)
			pF	ns	ns
Min	75	100			
Max	325	300	8	35	300

Limits	$\|h_{FE}\|$ V_{CE} = 10 V dc I_C = 20 mA dc f = 100 MHz	$V_{CE(sat)2}$ (1) I_C = 500 mA dc I_B = 50 mA dc	$V_{BE(sat)2}$ (1) I_C = 500 mA dc I_B = 50 mA dc
		V dc	V dc
Min	2.5		
Max	10.0	1.0	2.0

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for JANS devices in this specification sheet from lowest to highest are as follows: "M", "D", "P", "L", "R", "F", "G", and "H"). The RHA levels that are applicable for JANTXV devices in this specification sheet from lowest to highest are as follows: ("R", and "F").

1.5.3 Device type. The designation system for the device types of semiconductors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the semiconductors covered by this specification sheet are as follows: "6989", and "6990".

1.5.4 Suffix symbols. The following suffix symbols are incorporated in the PIN for this specification sheet:

	A blank second suffix symbol indicates a 14-pin dual-in-line (2N6989) in accordance with figure 1, or a 14-pin flat-pack (2N6990) in accordance with figure 2.
U	Indicates a 20-pin leadless chip carrier. (see figure 4).

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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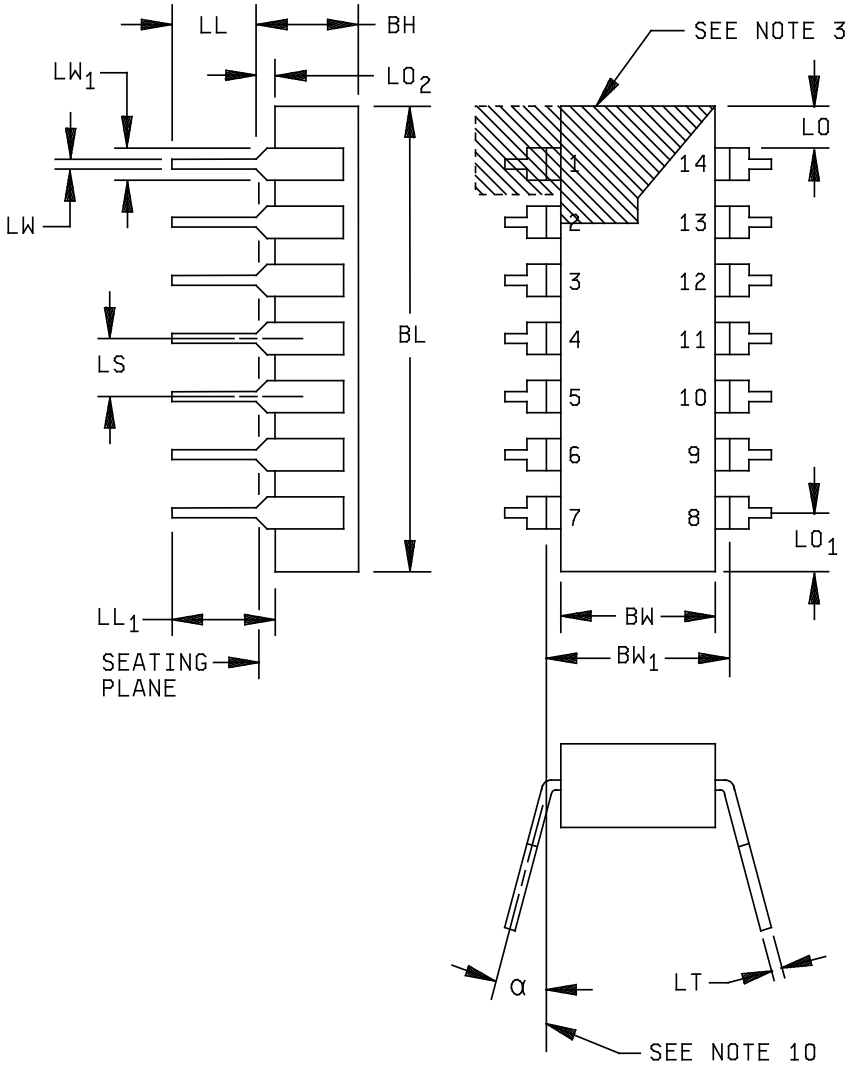


FIGURE 1. Dimensions and configuration for type 2N6989.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BH		.200		5.08	
LW	.014	.023	0.36	0.58	10
LW_1	.030	.070	0.76	1.78	4, 10
LT	.008	.015	0.20	0.38	10
BL		.785		19.94	6
BW	.220	.310	5.59	7.87	6
BW_1	.290	.320	7.37	8.13	9

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
LS	.100 BSC		2.54 BSC		7, 11
LL	.125	.200	3.18	5.08	
LL_1	.150		3.81		
LO	.005		0.13		8
LO_1		.098		2.49	8
LO_2	.015	.060	0.38	1.52	5
α	0°	15°	0°	15°	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Index area: A notch or pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
4. The minimum limit for dimension LW_1 may be .023 inch (0.58 mm) for leads number 1, 7, 8, and 14 only.
5. Dimension LO_2 shall be measured from the seating plane to the base plane.
6. This dimension allows for off-center lid, meniscus, and glass overrun.
7. The basic pin spacing is .100 inch (2.54 mm) between centerlines. Each pin centerline shall be located within $\pm$.010 inch (0.25 mm) of its exact longitudinal position relative to pins 1 and 14.
8. Applies to all four corners (leads number 1, 7, 8, and 14).
9. Lead center when α is 0 degrees. BW_1 shall be measured at the centerline of the leads.
10. All leads.
11. Twelve spaces.
12. No organic or polymeric materials shall be molded to the bottom of the package to cover the leads.
13. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Dimensions and configuration for type 2N6989 - Continued.

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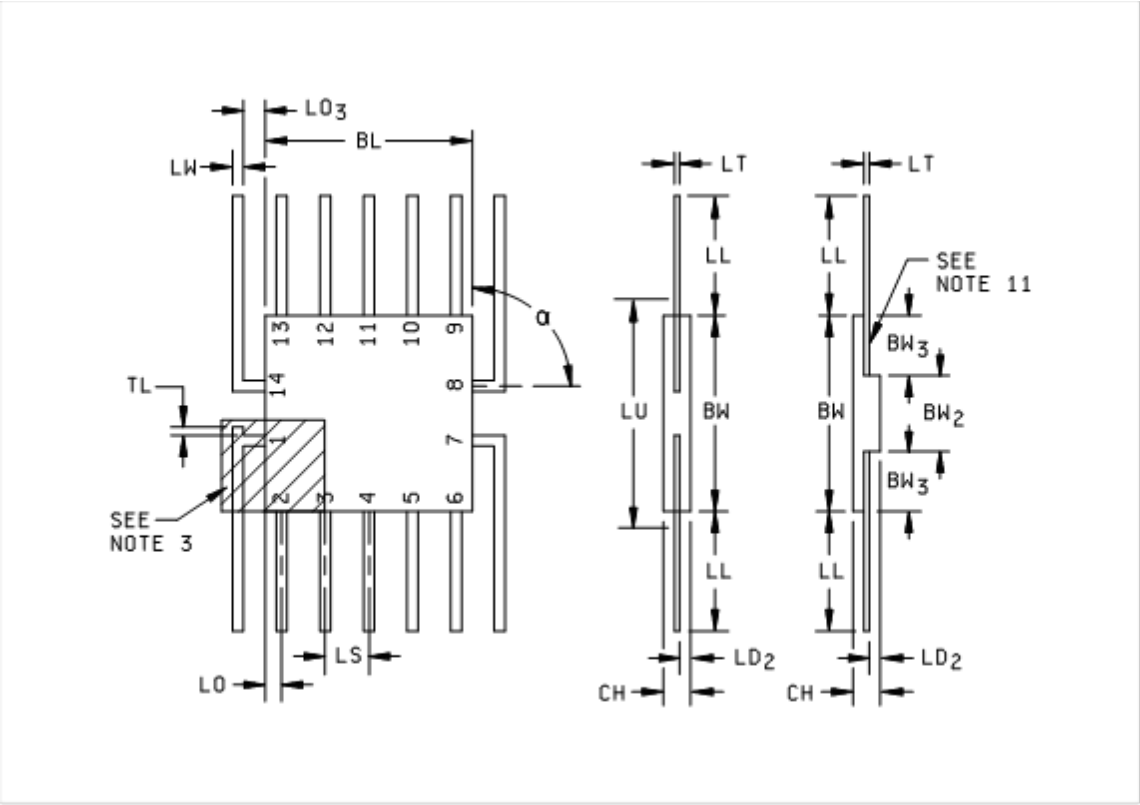


FIGURE 2. Physical dimensions for type 2N6990.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.030	.115	0.76	2.92	
LW	.010	.019	0.25	0.48	7
TL	.008	.015	0.20	0.38	12
BL		.280		7.11	5
BW	.240	.260	6.10	6.60	
LU		.290		7.37	5
BW_2	.125		3.18		

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BW_3	.030		0.76		
LS	.050 BSC		1.27 BSC		6, 8
LT	.003	.006	0.076	0.152	7
LL	.250	.370	6.35	9.40	
LD_2	.005	.040	0.13	1.02	4
LO	.005		0.13		9, 10
LO_3	.004		`0.10		13
α	30°	90°	30°	90°	14

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Index area: A notch or pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark. Alternatively, a tab (dim TL) may be used to identify pin one.
4. Dimension LD_2 shall be measured at the point of exit of the lead from the body.
5. This dimension allows for off-center lid, meniscus, and glass overrun.
6. The basic pin spacing is .050 inch (1.27 mm) between centerlines. Each pin centerline shall be located within ±.005 inch (0.13 mm) of its exact longitudinal position relative to pins 1 and 14.
7. All leads: Increase maximum limit by .003 inch (0.08 mm) measured at the center of the flat when the lead finish is solder.
8. Twelve spaces.
9. Applies to all four corners (leads number 2, 6, 9, and 13).
10. Dimension LO may be .000 inch (0.00 mm if leads number 2, 6, 9, and 13) bend toward the cavity of the package within one lead width from the point of entry of the lead into the body or if the leads are brazed to the metallized ceramic body.
* 11. No organic or polymeric materials shall be molded to the bottom of the package to cover the leads. This view shows one of two optional lead configurations available.
12. Optional, see note 1. If a pin one identification mark is used in addition to this tab, the minimum limit of dimension TL does not apply.
13. Applies to leads number 1, 7, 8, and 14.
14. Lead configuration is optional within dimension BW except dimensions LW and LT apply.
15. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
16. Pins 1, 7, 8, and 14 are collectors.
17. Pins 2, 6, 9, and 13 are bases.
18. Pins 3, 5, 10, and 12 are emitters.
19. Pins 4 and 11 are no contacts.

FIGURE 2. Physical dimensions for type 2N6990 - Continued.

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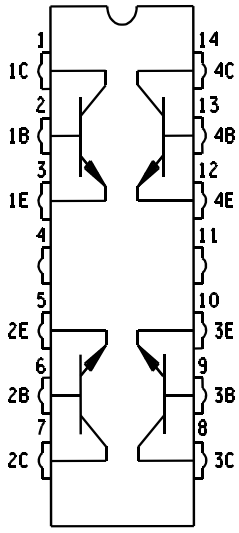
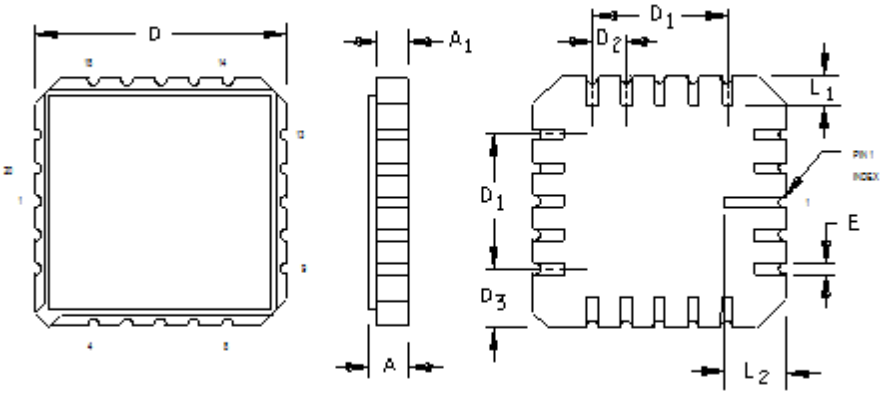


FIGURE 3. Schematic and terminal connections for type 2N6989 and 2N6990.

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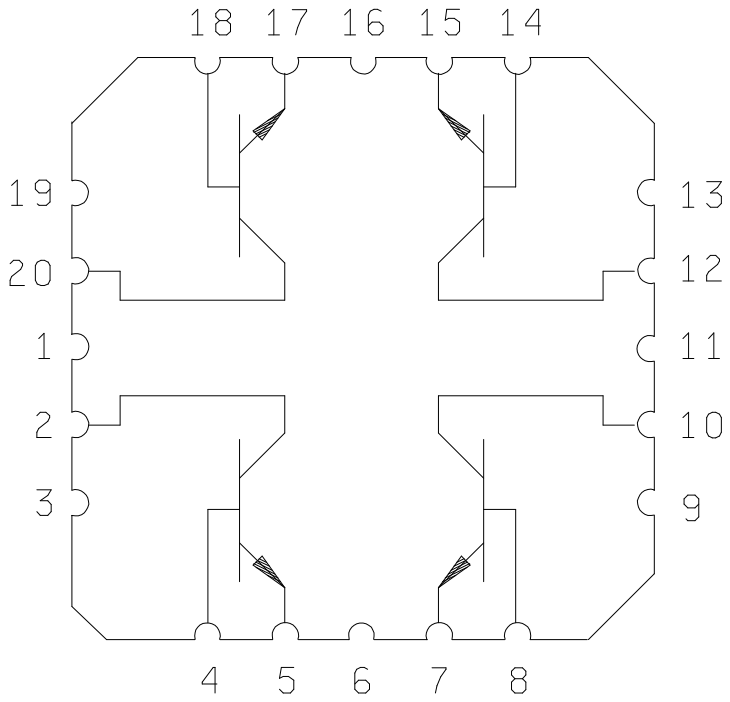


Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.073	.085	1.85	2.16
A_1	.063	.075	1.60	1.91
D	.345	.355	8.76	9.02
D_1	.195	.205	4.95	5.21
D_2	.050 TYP		1.27 TYP	
D_3	.070	.080	1.78	2.03
E	.025 REF		0.64 REF	
L_1	.050 REF for pins 2 through 20		1.27 REF for pins 2 through 20	
L_2	.080	.090	2.03	2.29

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 4. Physical dimensions for type 2N6989U.

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BOTTOM VIEW

	Pin numbers		
	Base	Collector	Emitter
First transistor	4	2	5
Second transistor	8	10	7
Third transistor	14	12	15
Fourth transistor	18	20	17

FIGURE 5. Schematic and terminal connections for type 2N6989U.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

DPA	Destructive physical analysis.
ESD	Electrostatic discharge.
PCB	Printed circuit board.
$R_{\theta JA}$	Thermal resistance junction to ambient.
$R_{\theta JA(AM)}$	Thermal resistance junction to ambient (adhesive mount to PCB).
$R_{\theta JC}$	Thermal resistance junction to case.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1, figure 2, figure 3, figure 4, and figure 5.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Schematic and terminal connections. The schematic and terminal connections shall be as shown on figure 3 (for flat package and dual-in-line) and on figure 5 (for leadless chip carrier).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500. The radiation hardened designator M, D, P, L, R, F, G, or H shall immediately precede (or replace) the device “2N” identifier (depending upon degree of abbreviation required).

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTX and JANTXV levels
(1) 3c	Thermal impedance, method 3131 of MIL-STD-750, see 4.3.2.	Thermal impedance, method 3131 of MIL-STD-750, see 4.3.2.
9	I_{CBO2}, h_{FE4}	Not applicable.
10	48 hours minimum.	48 hours minimum.
11	I_{CBO2}; h_{FE4}; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE4} = ±15 percent.	I_{CBO2}; h_{FE4}.
12	See 4.3.1.	See 4.3.1.
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15 percent.	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15 percent.

(1) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed on each die in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). See table III, group E, subgroup 4 herein. (See figure 10, figure 11, figure 12, and figure 13.)

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of table I, group A1 and A2 inspection only (table E-VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Delta requirements shall be in accordance with 4.5.5 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with table I, subgroup 2 and 4.5.5 herein.

4.4.2.1 Group B inspection, table (JANS) E-VIa of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.
B5	1027	(NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) V_{CB} = 10 V dc, $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). Option 1: 96 hours minimum sample size in accordance with table E-VIa of MIL-PRF-19500, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve T_J = +225°C minimum.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot and wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Delta requirements shall be in accordance with 4.5.5 herein, delta parameters apply to subgroup C6.

4.4.3.1 Group C inspection, table E-VII (JANS) of MIL-PRF-19500.

	Subgroup	Method	Condition
*	C2	2036	Test condition E, 3 ounce (85 Grams) weight; three bends of 15 degrees for 2N6990; three bends for 2N6989; not applicable to 2N6989U.
	C6	1026	1,000 hours at V_{CB} = 10 V dc; T_J = +150°C min.

4.4.3.2 Group C inspection, table E-VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

	Subgroup	Method	Condition
*	C2	2036	Test condition E, 3 ounce (85 Grams) weight; three bends of 15 degrees for 2N6990; three bends for 2N6989; not applicable to 2N6989U.
	C5	3131	$R_{\theta JA}$, see 1.3.
	C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2.e herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified table III herein. Electrical measurements (end-points) and delta measurements shall be in accordance with the applicable steps of table I, subgroup 2 and 4.5.5 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

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4.5.3 Independent transistor inspections. Inspections shall be performed on each transistor in the array.

4.5.4 Transistor-to-transistor resistance. The leads of each transistor shall be shorted together for this test. The resistance shall be measured between each transistor in the array.

4.5.5 Delta requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1	Collector-base cutoff current.	3036	Bias condition D, V_{CB} = 60 V dc.	ΔI_{CB02} (1)	100 percent of initial value or 8 nA dc, whichever is greater.	
2	Forward current transfer ratio.	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed see 4.5.1.	Δh_{FE4} (1)	±25 percent change from initial reading.	

(1) Devices which exceed the table I limits for this test shall not be accepted.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071					
Solderability 3/ 4/ 5/	2026					
Resistance to solvents 3/ 4/ 5/	1022					
Temp cycling 3/ 4/	1051					
Heremetic seal 4/	1071					
Fine leak Gross leak						
Electrical measurements 4/						
Bond strength 3/ 4/	2037					
Decap internal visual design verification 4/	2075					
Subgroup 2						
Thermal impedance 6/	3131	See 4.3.2.	$Z_{\theta JX}$			°C/W
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 75 V dc I_C = 10 µA dc.	I_{CBO1}		10	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6 V dc I_E = 10 µA dc.	I_{EBO1}		10	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1).	$V_{(BR)CEO}$	50		V dc
Collector to base cutoff Current	3036	Bias condition D; V_{CB} = 60 V dc.	I_{CBO2}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc.	I_{EBO2}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc.	h_{FE1}	50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc.	h_{FE2}	75	325	

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc.	h_{FE3}	100		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1).	h_{FE4}	100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1).	h_{FE5}	30		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1).	$V_{CE(sat)1}$		0.3	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1).	$V_{CE(sat)2}$		1.0	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1).	$V_{BE(sat)1}$	0.6	1.2	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1).	$V_{BE(sat)2}$		2.0	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 60 V dc.	I_{CBO3}		10	µA dc
Low temperature operation		T_A = -55°C.				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc.	h_{FE6}	35		
Subgroup 4						
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz.	h_{fe}	50		
Magnitude of small-signal short- circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 20 mA dc; f = 100 MHz.	$\|h_{fe}\|$	2.5	10.0	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz.	C_{obo}		8	pF

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Input capacitance (output open- circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz (see 4.5.2).	C_{ibo}		25	pF
Turn-on time		(See figure 14)	t_{on}		35	ns
Turn-off time		(See figure 15)	t_{off}		300	ns
Transistor-to-transistor resistance		| V_{T-T} | = 500 V dc; (see 4.5.4).	R_{T-T}	10^{10}		Ω
Subgroups 5 and 6						
Not applicable						

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
* 6/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroup 3, 4, and 5 (JANS).
Group B, step 1 (TX and TXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1 and 2.

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TABLE II. Group D inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 3/						
Neutron irradiation	1017	Neutron exposure V_{CES} = 0 V				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 75 V dc	I_{CBO1}		20	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6 V dc	I_{EBO1}		20	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	50		V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		20	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO2}		20	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	[h_{FE1}] 4/	[25]		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	[h_{FE2}] 4/	[37.5]	325	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	[h_{FE3}] 4/	[50]		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	[h_{FE4}] 4/	[50]	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	[h_{FE5}] 4/	[15]		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		.35	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.15	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.6	1.38	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.3	V dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2						
Total dose irradiation	1019	Gamma exposure V_{CES} = 40 V Condition A				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 75 V dc	I_{CBO1}		20	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6 V dc	I_{EBO1}		20	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	50		V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		20	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO2}		20	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	[h_{FE1}] 4/	[25]		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	[h_{FE2}] 4/	[37.5]	325	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	[h_{FE3}] 4/	[50]		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	[h_{FE4}] 4/	[50]	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	[h_{FE5}] 4/	[15]		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		.35	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; (see 4.5.1)	$V_{CE(sat)2}$		1.15	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.6	1.38	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.3	V dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

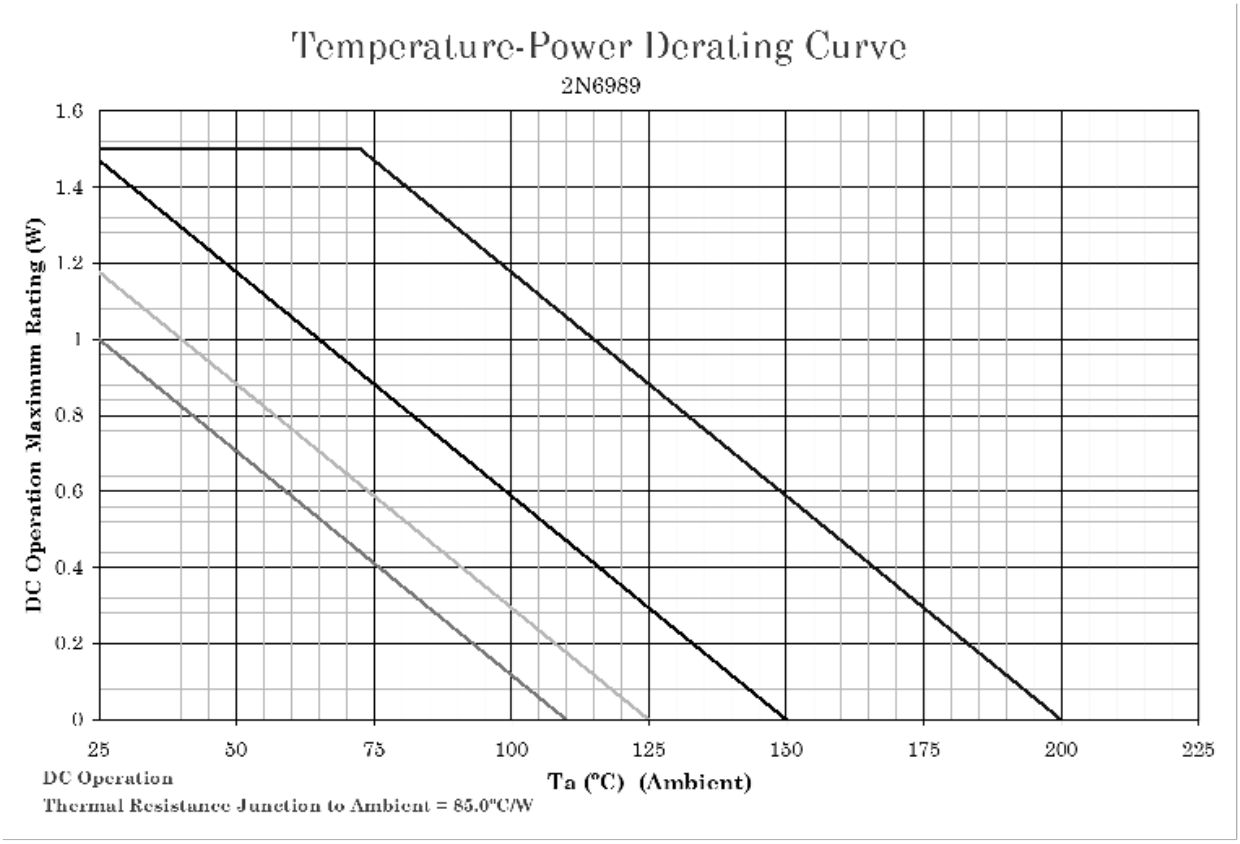
1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ See 6.2.e herein.
4/ See method 1019 of MIL-STD-750 for how to determine [h_{FE}] by first calculating the delta ($1/h_{FE}$) from the pre- and post-radiation h_{FE}. Notice the [h_{FE}] is not the same as h_{FE} and cannot be measured directly. The [h_{FE}] value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Group E inspection (all quality levels) - for qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	45 devices c = 0
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and 4.5.5 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	V_{CB} = 10 - 30 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of +100°C.	
Electrical measurements		See table I, subgroup 2 and 4.5.5 herein.	
Subgroup 4			
Thermal impedance curves	3131	See MIL-PRF-19500.	Sample size N/A
Subgroup 5			
Not applicable			
Subgroups 6			11 devices
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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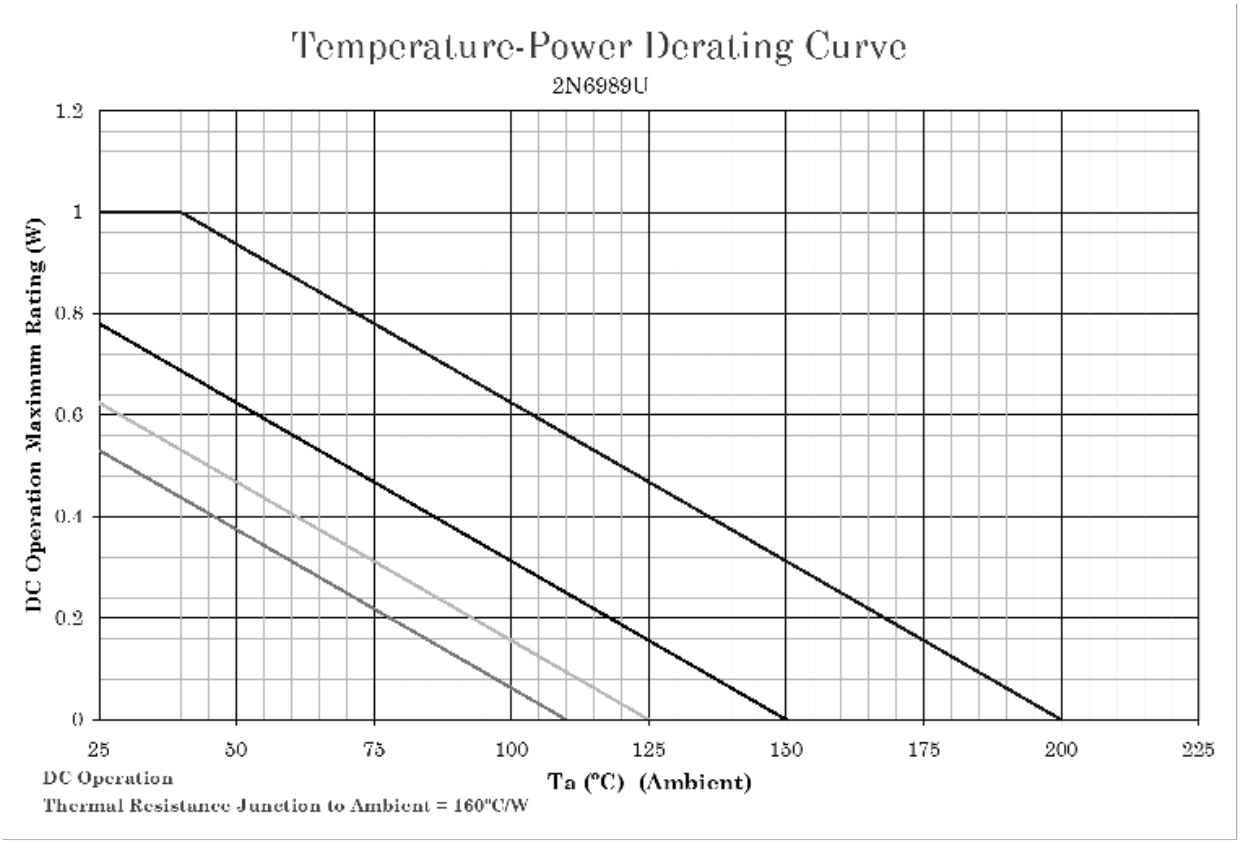


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 6. Derating for 2N6989, ($R_{\theta JA}$).

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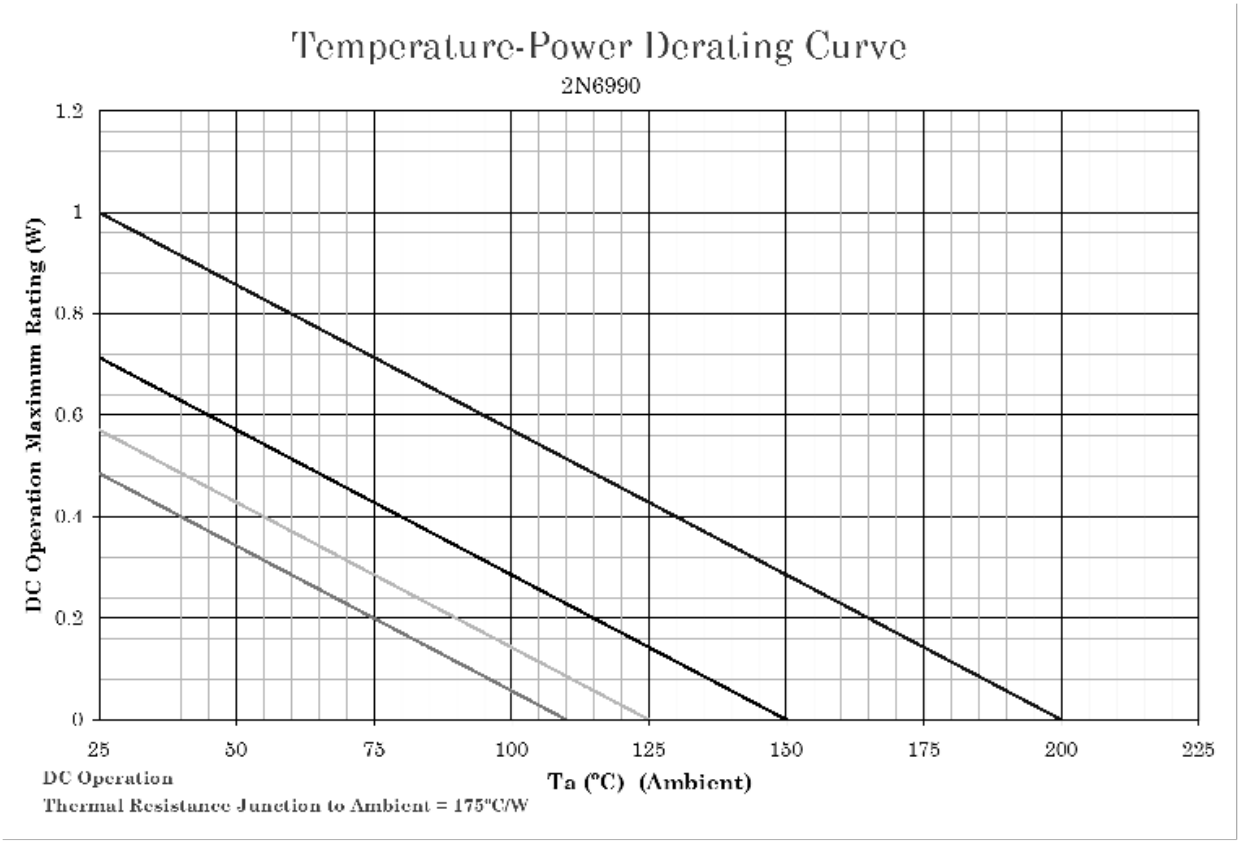


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 7. Derating for 2N6989U, ($R_{\theta JA}$).

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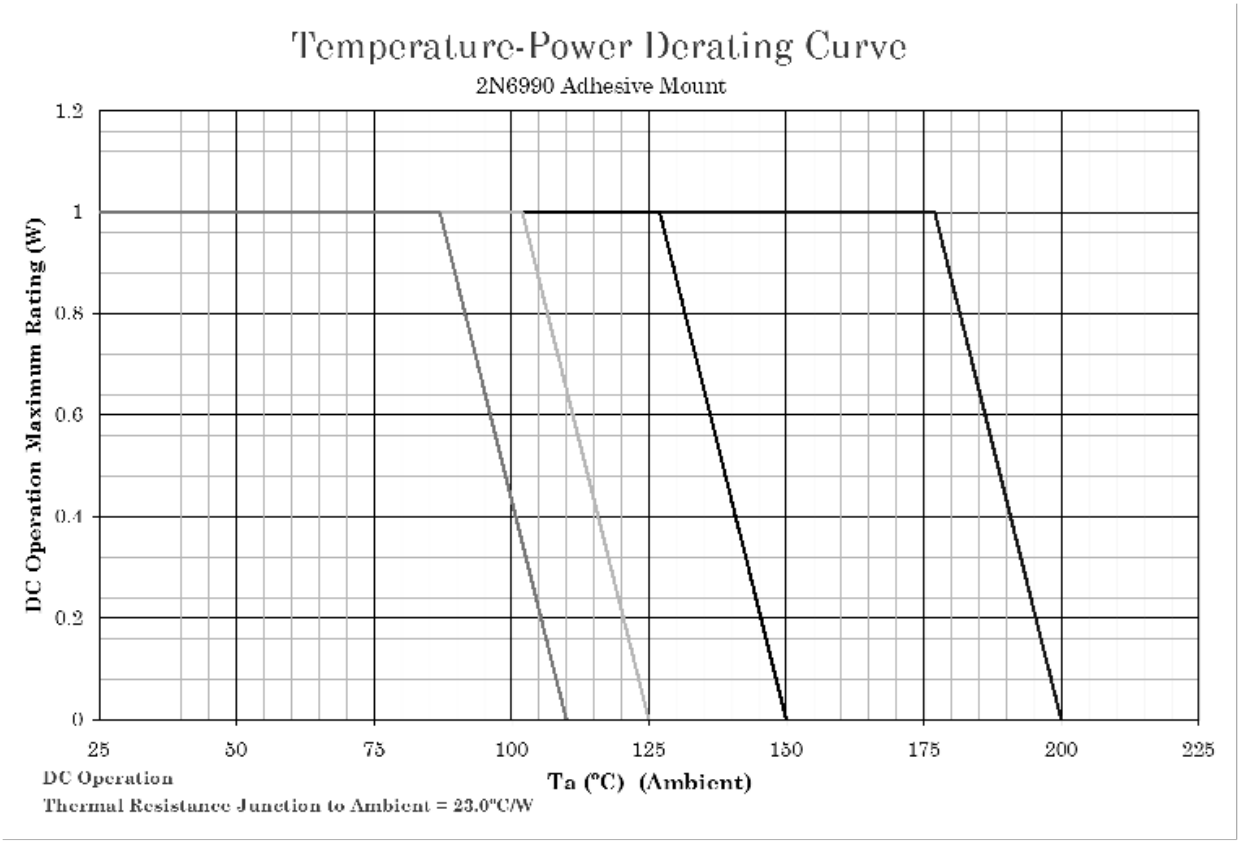


NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 8. Derating for 2N6990, ($R_{\theta JA}$).

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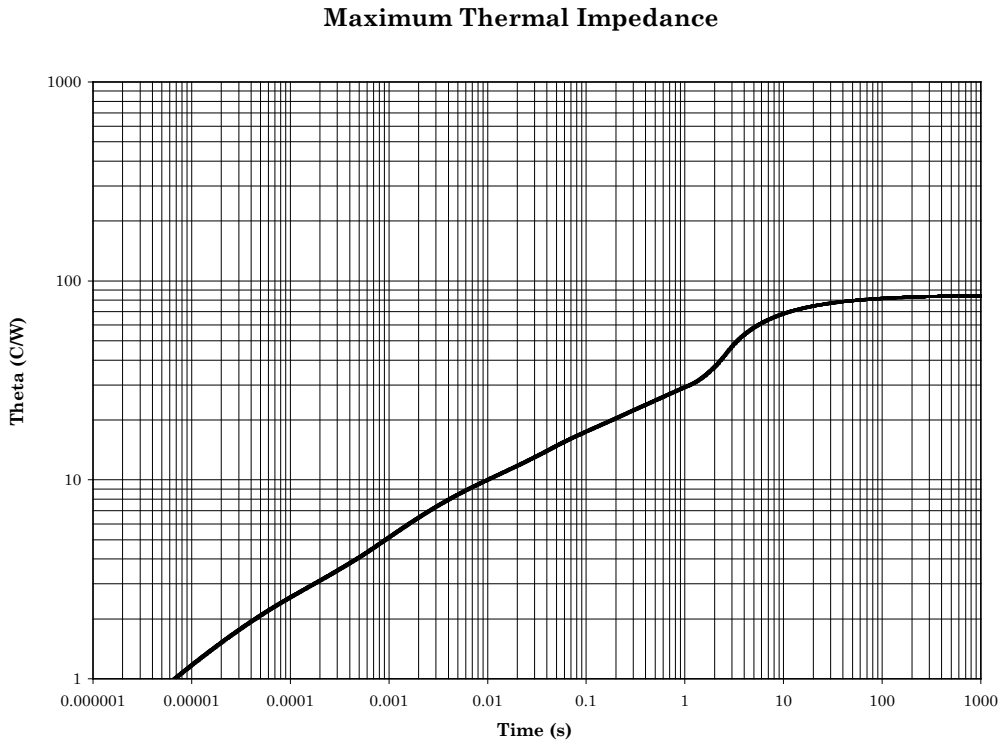


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 9. Derating for 2N6990, ($R_{\theta JA}$ adhesive mount).

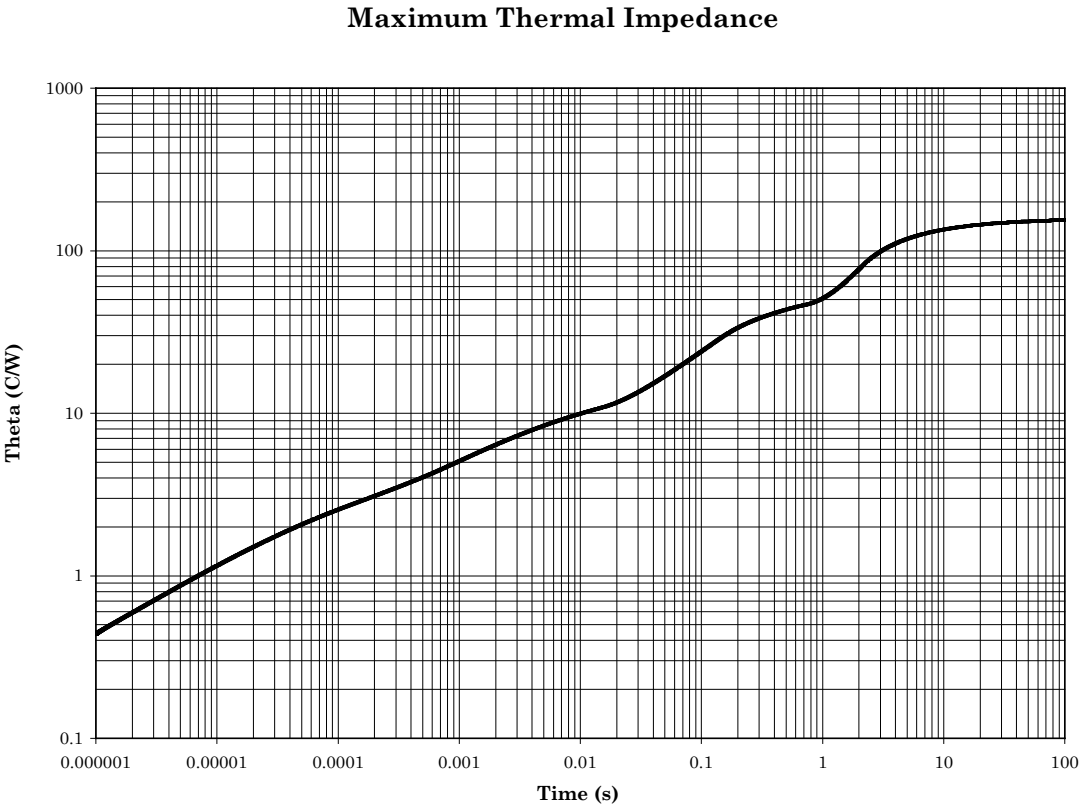
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NOTE: For response curve characteristics for a single chip while other chips are powered identically but independently, multiply the vertical axis thermal impedance values by 4.

FIGURE 10. Thermal impedance graph ($R_{\theta JA}$) for 2N6989 (DIP14).

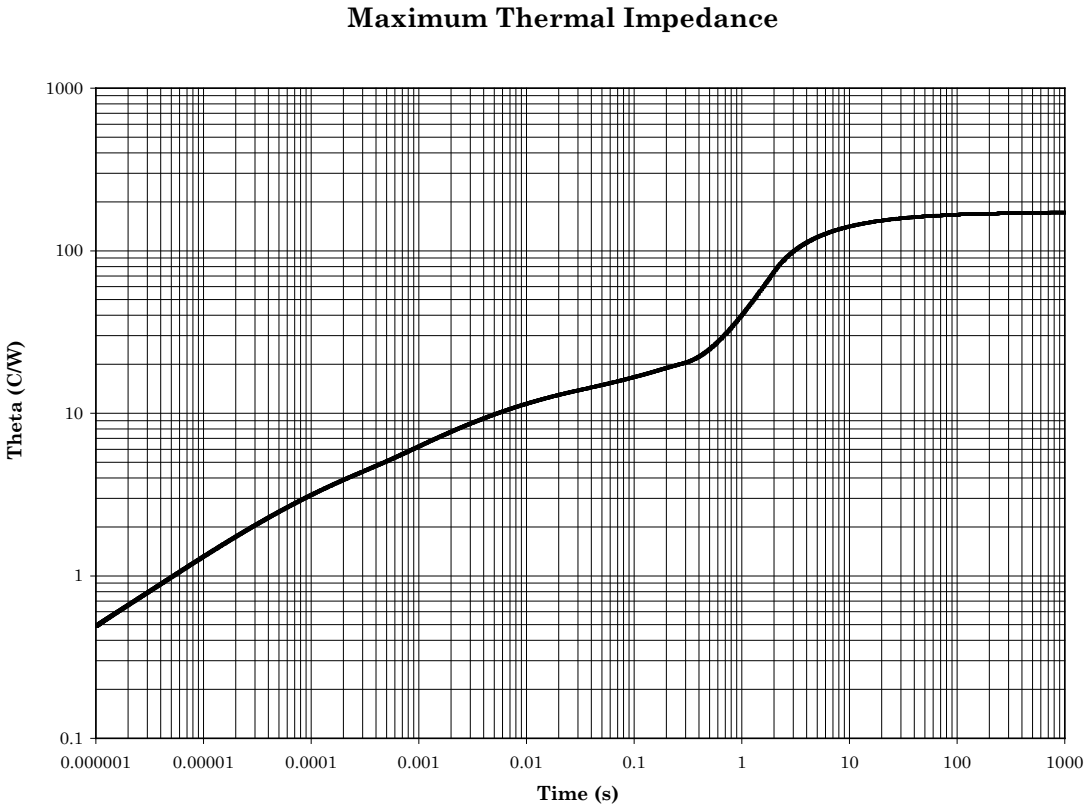
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NOTE: For response curve characteristics for a single chip while other chips are powered identically but independently, multiply the vertical axis thermal impedance values by 4.

FIGURE 11. Thermal impedance graph ($R_{\theta JA}$) for 2N6989U (20 pin leadless chip).

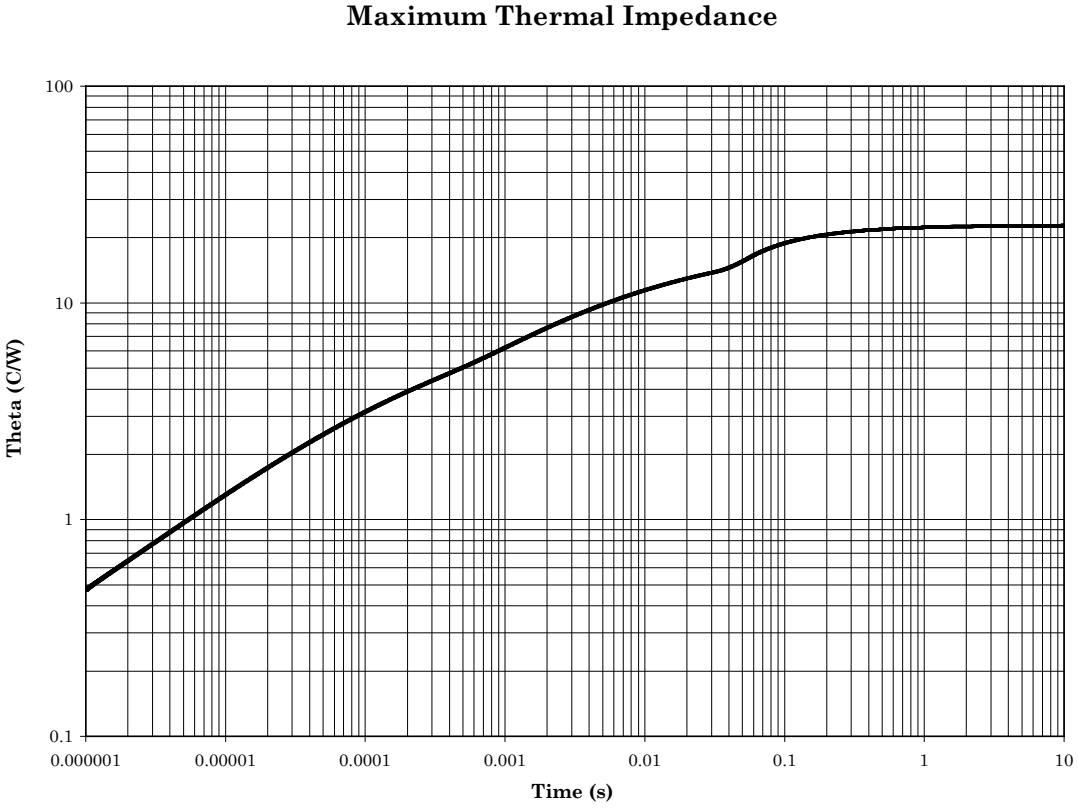
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NOTE: For response curve characteristics for a single chip while other chips are powered identically but independently, multiply the vertical axis thermal impedance values by 4.

FIGURE 12. Thermal impedance graph ($R_{\theta JA}$) for 2N6990 (14 pin flat pack).

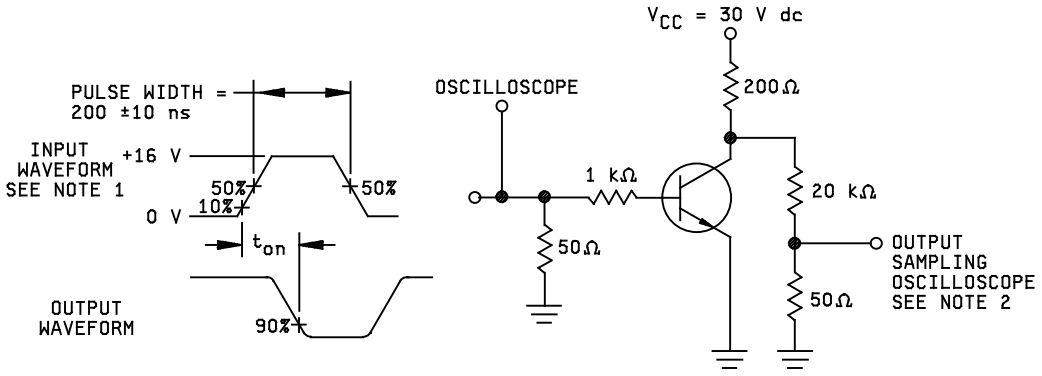
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NOTE: For response curve characteristics for a single chip while other chips are powered identically but independently, multiply the vertical axis thermal impedance values by 4.

FIGURE 13. Thermal impedance graph ($R_{\theta JA}$)AM for 2N6990 (14 pin flat pack adhesive mount).

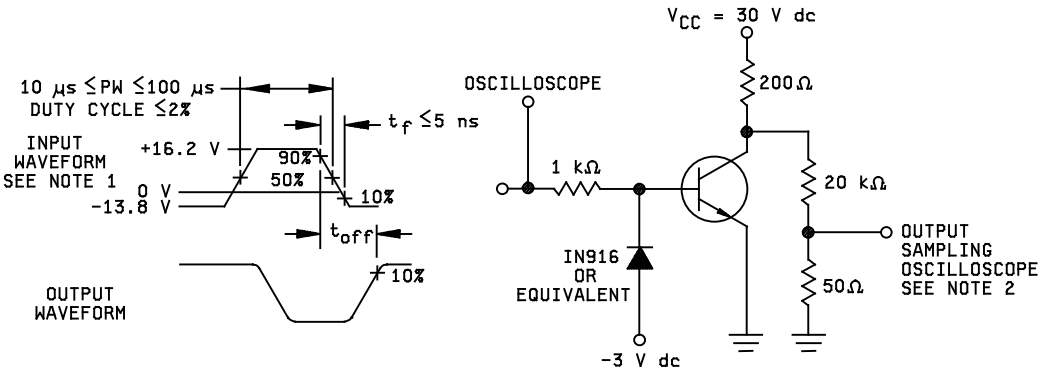
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NOTES:

1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 14. Saturated turn-on switching time test circuit.



NOTES:

1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 15. Saturated turn-off switching time test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

e. The complete Part or Identifying Number (PIN), see 1.5 and 6.4.

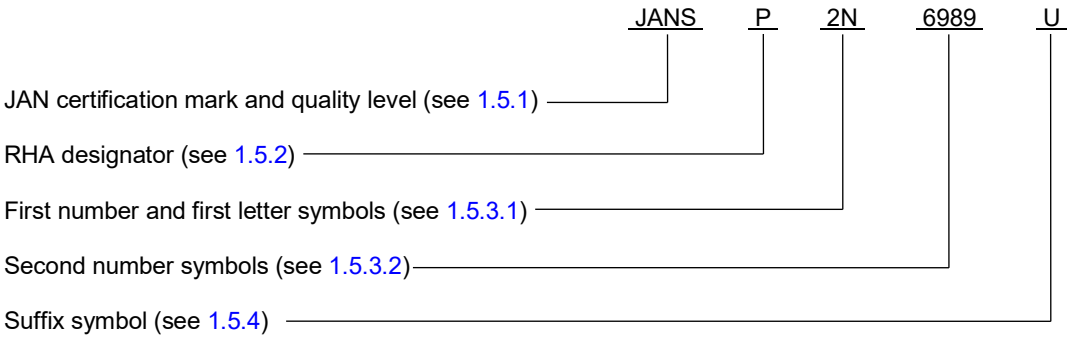
f. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it must be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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6.4 PIN construction example.

6.4.1 Encapsulated devices. The PINs for encapsulated devices are constructed using the following form.



6.5 List of PINs.

6.5.1 PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for types 2N6989 (1)	PINs for types 2N6990 (1)
JAN2N6989	JAN2N6990
JAN2N6989U	JANTX2N6990
JANTX2N6989	JANTXV2N6990
JANTX2N6989U	JANTXVR2N6990
JANTXV2N6989	JANTXVF2N6990
JANTXV2N6989U	JANS2N6990
JANTXVR2N6989	JANS#2N6990
JANTXVF2N6989	
JANTXVR2N6989U	
JANTXVF2N6989U	
JANS2N6989	
JANS2N6989U	
JANS#2N6989	
JANS#2N6989U	

(1) The number sign (#) represents one of eight RHA designators available (M, D, P, L, R, F, G, or H).

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6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
* Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2020-089)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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